

React HandsOn 14: Context API Employee Management Theme

Objectives

Explain the need and Benefits of React Context API

Working with createContext()

List the types of Router Components

In this hands-on lab, you will learn how to:

Create a context to be used by child components

Create a provider and consumer of the context

Prerequisites

The following is required to complete this hands-on lab:

Node.js

NPM

Visual Studio Code

Notes

Estimated time to complete this lab: 30 minutes.

Developers of Apps Centric Solutions have created an employee management application which supports light and dark themes for the buttons. The current solution uses the react state and props to provide the theme name to be used from App component to Employee List component and from there to Employee Card component. Quality assurance team analyzed the solutions and found the technique being used to be a substandard one. React architect suggested to use the react context API to share the theme name with nested child components instead of passing them down using props from the parent component.

You are assigned the task of converting the application from props only to React Context API.

Application can be downloaded from below

Unzip the application and open it using VS Code

Go to terminal and execute npm install command to restore all the node modules

Figure 1: Restore node modules

Run the application once to see the output. Use npm start command.

Figure 2: Starting application

Explore the components present in App.js, EmployeesList.js and EmployeeCard.js files.

Create a new file with the name as ThemeContext.js. Define a new context in the file with the name as ThemeContext and assign it a default value of 'light' and export it as default from the module.

Open App component present in App.js file.

Import the ThemeContext in App component.

Define the theme context provider to be the entire JSX of the App component.

Assign the value for the theme provider from the state of the component.

Modify the call to EmployeeList component so that theme name is no longer passed as props.

Go to EmployeeList component present in EmployeeList.js file and modify it so that theme name is not passed

React HandsOn 14: Context API Employee Management Theme

explicitly to its child component.

Go to EmployeeCard component inside EmployeeCard.js file

Import the ThemeContext into the component file

Retrieve the value of the context with the help of useContext() and store it in a variable

Use the variable to pass the className for the buttons.